Breast Cancer Gene Expression Analysis Using Bioinformatics

Abstract
The present study differentiates breast tumor and normal tissue sample from the organism Homo sapiens , dataset GSE139038 using bioinformatics to understand its molecular mechanism and biological significance of the identified gene. Some methods which helped were Differential gene expression analysis, data visualization, Functional enrichment analysis , biological interpretation . The study provides significant insight into the potential role of identified genes and their importance as a biomarker and their molecular mechanism.

Keywords
Breast cancer, Differential gene expression , GEO, Enricher,Reactome pathway , Gene Ontology , Bioinformatics , Volcano plot , heatmap

1. INTRODUCTION

Breast cancer occurs when breast cells mutate and become cancerous cells that multiply and form tumors. It is the most frequently diagnosed cancer globally, primarily affecting women of age 50 or older , but can also affect men or younger women.

This disease is important because early detection through regular screenings can increase survival rate. Since most individuals diagnosed have no family history, widespread awareness is essential. This allows people to recognize early symptoms, understand genetic factors, and adopt healthy lifestyles that decrease their overall risk. Therefore, understanding the molecular mechanisms involved in breast cancer is important for improving knowledge of disease and identifying potential biomarkers and therapeutic targets .

Although every cell contains the same DNA , different genes are switched on or off depending on the type and condition of the cell. In breast cancer, changes in gene expression can affect synthesization of a biological product like cell growth , cell division and cell death . So, by comparing gene expression between breast tissue and normal tissue, researchers can identify differentially expressed genes. Studying these genes helps us understand the molecular change involved in breast cancer .

The availability of public databases has made it easier for researchers to study disease using gene expression data . The most widely used public database provider is Gene expression omnibus (GEO) which provides free gene expression dataset from different biological studies . To compare gene expression between breast cancer and normal tissue samples,Bioinformatics tools help us identify differentially expressed genes and help researchers understand the biological process and pathways associated with the particular diseases. Functional enrichment analysis further helps us understand how these genes contribute to breast cancer development and progress.

Based on this, the study aims to identify differentially expressed genes associated with breast cancer using publicly available GEO dataset GSE139038. Different gene expression analysis was performed to compare breast tumor and normal tissue samples. The identified genes were further analyzed by using gene ontology and reactomse pathway Enrichment analysis, and the results were visualized using volcano plot and heatmap to understand the molecular mechanism of breast cancer in a better way.

2. Materials and Methods

2.1 Study Design

This study was as in the silico bioinformatics study which was conducted by publicly available gene expression data provided by Gene expression omnibus (GEO) to investigate molecular changes associated with breast cancer. The main objective was to identify differentially expressed genes(DEGs) between normal tissue samples and breast tumors and to further understand their biological significance . Differential gene expression analysis, data visThis study was an visualization and functional enrichment analysis were performed to have better insight into molecular mechanisms involved in breast cancer.

2.2 Data Acquisition.

Gene expression data that were used in this study were obtained from the National Center for Biotechnology Information(NCBI) Gene Expression Omnibus(GEO) database. The publicly available dataset GSE139038 was selected for the analysis. This dataset contains gene expression profile of Homo Sapienes breast tumor and normal breast tissue sample. The dataset contains 32 breast tumor samples and 33 normal tissue samples making it a total of 65 samples for study . This dataset was used specifically for study because it provides suitable transcriptomic data for comparing breast cancer tissue and normal tissue samples.

2.3 Differential Gene Expression Analysis

Differential gene expression analysis was performed using the online tool GEO2R to compare gene expression between breast tumor and normal tissue samples. The GEO2R tool uses statistical methods, specifically the limma package which applies linear models to identify statistically significant differences in gene expression , to identify genes that show significant differences in expression between the two groups. Genes with adjective p-value of equal to or less than 0.05 and an absolute log fold change of equal to or greater than 1 were considered significantly differentially expressed.

The significant genes were then classified as upregulated or downregulated on basis of their expression level in breast tumor sample compared with normal tissue

2.4 Data Visualization

Data visualization was performed to represent the differential gene expression patterns identified between breast tumor and normal tissue samples. The processed gene expression data and differential expression results were organized and analyzed using python based computational approaches.

The pandas library was used for data handling, organization and preprocessing of gene expression datasets. The matplotlib library was used to generate graphical representations, including volcano plot and heatmaps to visualize the distribution of differentially expressed genes and expression patterns across the analyzed samples.

The volcano plot was used to visualize the relationship between statistical significance and magnitude of gene expression changes, allowing the identification of significantly upregulated and downregulated genes. Heatmap visualization was performed to demonstrate differences in expression patterns among selected differentially expressed genes between breast tumor and normal tissue samples.

2.5 Functional Enrichment Analysis

Functional enrichment analysis was performed to discover the biological significance of the identified differentially expressed genes. All statistically significant DEGs obtained from the differential gene expression analysis were combined and subjected to enrichment analysis using the enricher online tool.

Gene ontology(GO) is a biological process enrichment analysis performed to identify biological functions and cellular processes associated with the identified DEGs. In addition, Reactomse pathway enrichment analysis was conducted to determine the molecular pathways potentially involved in breast cancer development and progression.

The enriched biological processes and pathways were analyzed based on their statistical significance to understand the functional roles of genes altered between breast tumor and normal tissue samples.

2.6 Candidate Gene Selection and Biological Interpretation

Following differential gene expression analysis, candidate genes were selected for further biological interpretation based on their statistical significance,magnitude of differential expressed genes and their significance in breast cancer. The top differentially genes were reviewed to identify those with significant roles in cancer related biological processes and molecular pathways.

The top 5 candidate genes were selected for in depth interpretation using published literature . The biological functions of these genes were then reported. The report involves the significance of genes in breast cancer, and their potential involvement as biomarkers or therapeutic targets were examined to support the interpretations of the bioinformatics findings.

3. Results

3.1 Identification of Differentially Expressed Genes

Differential gene expression analysis was performed to compare breast tumor samples with normal breast tissue samples using the predefined criteria of adjusted P- value < 0.05 and absolute log fold change(logFC) > 1. A total of 2300 differentially expressed genes were identified, indicating a prominent difference in gene expression between the two samples.

To identify the most prominently altered genes, the DEGs were ranked according to their log fold change. It was then that the top 10 upregulated genes and top 10 downregulated genes were selected. The most highly upregulated genes and downregulated genes are shown in Table 1 and Table 2

Table 1. Top 10 upregulated genes identified in breast tumor sample

Gene symbol	Adj.P.Val	logFC
LEP	8.890000e-16	5.673
MME	4.300000e-14	3.637
LEPR	1.170000e-13	3.569
ADATMS5	1.170000e-13	3.127
TUSC5	1.520000e-13	4.449
CFD	6.440000e-13	3.876
CHRDL1	1.560000e-12	3.952
CAV1	1.720000e-12	3.271
IGFBP6	2.270000e-12	3.258

DMN	2.430000e-12	3.734

Table 2. Top 10 downregulated genes identified in breast tumor sample

Gene symbol	adj.P.Val	logFC
C22orf18	2.530000e-09	-2.020
UBE2T	5.840000e-09	-2.501
COL11A1	6.520000e-09	-4.04
ESCO2	7.480000e-09	-2.43
BUB1B	1.450000e-08	-2.32
PTTG1	2.000000e-08	-2.13
PTTG1	2.870000e-08	-2.28
MELK	3.020000e-08	-2.59
KIAA0101	3.180000e-08	-2.39
C20orf129	8.310000e-08	-2.64

These genes showed the greatest differences between breast tumor samples and normal tissue samples and were also selected for biological interpretation

3.2 Visualization Of Gene Expression

Visualization of differentially gene expression patterns between breast tumor and normal tissue samples was generated by volcano plot and heatmap. The volcano plot provides an overview of the distribution of differentially expressed genes based on their statistical significance and log2 fold change. The upregulated genes were observed on the right side of the plot , whereas downregulated genes were observed on the left side. The volcano plot described a clear distinction between upregulated genes and downregulated genes , highlighting the overall changes in gene expression associated with breast cancer.

A heatmap was generated using the top 10 upregulated genes as well as top 10 downregulated genes to visualize their expression patterns across samples. The heatmap shows the difference in the expression profile between breast tumor and normal tissue samples . Upregulated genes show a higher expression level in tumor samples, compared to downregulated genes which show reduced expression opposite to that of normal breast tissue . These findings indicate that the selected genes effectively distinguish breast tumor sample to that of normal tissue sample and may be used as significant biomarkers for breast cancer

Figure 1 . Volcano plot showing differentially expressed genes between breast tumor and normal tissue samples

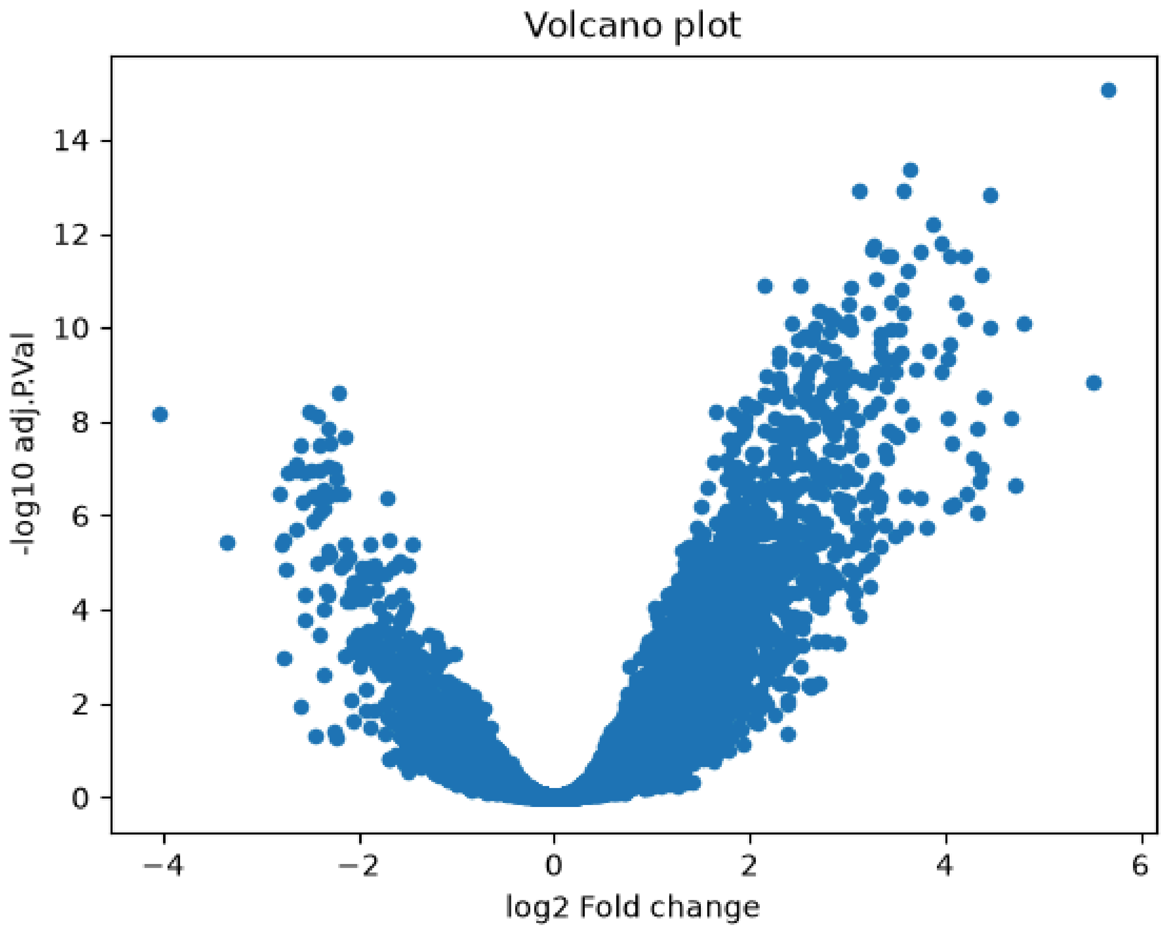
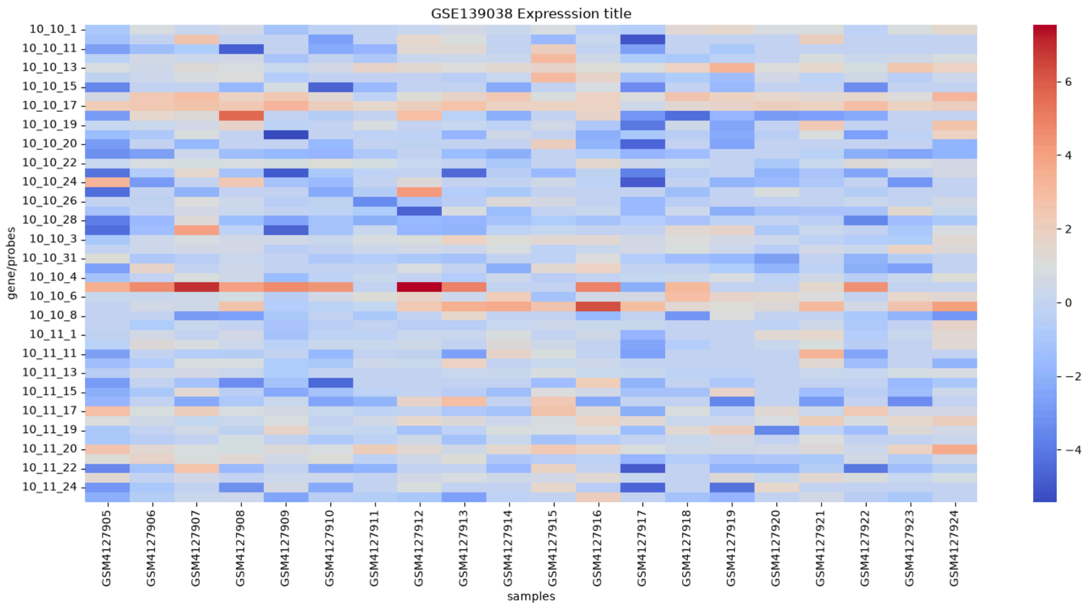


Figure 2 . Heatmap showing top 10 upregulated genes and top 10 downregulated genes.



3.3 Functional Enrichment Analysis

3.3.1 Gene Ontology(GO) Biological Process Analysis

To define the biological functions associated with identified differentially expressed genes , Gene ontology biological process enrichment analysis was performed using an enricher platform . The analysis revealed enrichment of several biological processes related to cell cycle regulation , mitotic spindle attachment , cytokinesis , cell division , DNA recombination and interferon mediated cellular response . These biological processes are closely related to cellular proliferation, chromosome segregation, immune response and genome stability , all of which play significant roles in development and progression of breast cancer.

Among the enriched GO biological processes , positive regulation of cell cycle processes , regulation of attachment of spindle microtubules to kinetochore, regulation of cytokinesis , and positive regulation of cell division showed the strongest enrichment based on the values . Additional processes included cellular response to interferon beta , cellular response to alkaloids, regulation of DNA recombination and subtelomeric heterochromatin formation , suggesting alteration in immune signalling and genomic maintenance in breast cancer.

Table 3. Top 10 enriched Gene Ontology terms

Rank	Gene Ontology Biological processes	P-value
1	Positive regulation of cell cycle process	2.6 x 10-5
2	Regulation of attachment of spindle microtubule To kinetochore	8.6 x 10-5
3	Regulation of cytokinesis	2.3 x 10-4
4	Mitotic signaling pathway	2.8 x 10-4
5	Positive regulation of cytokinesis	5.5 x 10-4
6	Positive regulation of cell division	9.7 x 10-4
7	Cellular response to interferon-beta	1.65 x 10-3
8	Cellular response to alkaloid	2.6 x 10-3
9	Regulation of DNA recombination	3.08 x 10-3
10	Subtelometric heterochromatin formation	3.2 x10-3

3.3.2 Reactomase Pathway Analysis

Reactomase pathway enrichment analysis further identified several significant enriched pathways related to homologous recombination repair, DNA double strand break repair, DNA strand elongation, and interferon gamma signaling . They indicate that slight alterations or damage in DNA repair mechanisms and cell cycle regulation play an important role in molecular pathogenesis of breast cancer. Additional enriched processes included Activation of pre- replicative complex, lagging strand synthesis, Impaired BRCA2 binding to PALB2, Resolution of D loop structures through holliday junction intermediates, suggesting alterations in DNA and genomic maintenance in breast cancer

Rank	Reactomase Pathway	P-Value
1	Defective homologous recombination repair Due to BRCA2 loss of function	5.457 x 10-4
2	Disease of DNA double strand break repair	5.457 x 10-4
3	Homologous DNA pairing and strand exchange	7.536x 10-4
4	Interferon Gamma signaling	7.805 x 10-4
5	DNA strand elongation	8.122 x 10-4
6	Activation of the pre- Replicative complex	9.632 x 10-4
7	Telomere C- strand synthesis	1.135 x 10-3

8	Impaired BRCA2 binding to RAD51	1.329 x 10-3
9	Resolution of D-loop structures through Holliday junction intermediates	1.329 x 10-3
10	Impaired BRCA2 binding to PALB2	1.356 x 10-3

3.3 Biological Interpretation Of Key Genes

The top differentially expressed genes identified from the dataset GSE139038 were examined to understand their biological significance and potential involvement in breast cancer progression.

1. **LEP**

   LEP encodes for leptin, a hormone involved in energy regulation,immune signalling. In breast cancer, altered expression may influence tumor growth by promoting inflammation , cell proliferation and survival pathways. It also serves as a regulator of weight , obesity and energy homeostasis.

2. **MMN**

   MMN encodes for Neprilysin which is a cell surface enzyme that regulates Signalling molecules by breaking them. In breast cancer, Changes in MME Expression may result in its effect on cellular communication , tumor Interactions and pathways involved in cancer growth and invasion.

3. **LEPR**

   Leptin Receptor allows cells to respond to leptin signals. Alteration in LEPR expression may enhance leptin mediated pathways that promote tumor cell proliferation, survival and metastatic potential in breast cancer.

4. **ADAMTS5 Family Gene**

   ADAMTS5 proteins consist of multi-domain enzyme genes in humans that break down proteins outside cells. Dysregulation of these genes may contribute to tissue remodelling during cancer progression.

5. **TUSC5**

   TUSC5 which is a Tumor Suppression Candidate genes are involved in Maintaining normal cellular control and preventing abnormal growth. Reduced TUSC5 expression may weaken tumor suppressive mechanism, Allowing uncontrolled cell proliferation.

6. **UBE2T**

Ubiquitin Conjugating Enzyme E2 T is involved in DNA repair and genome Genome stability. In breast cancer, Alteration in UBE2T may affect DNA Damage response, allowing accumulation of genetic errors and supporting Cancer cell survival

7. **COL11A1**

Collagen Type X1 Alpha 1 Chain helps build supportive framework around Cells. In cancer, increased COL11A1 expression may create a more Favourable environment for tumor cells by supporting its movement, Invasion and interaction with other tissue

8. **ESCO2**

Establishment of Cohesion 1 Homolog 2 helps keep newly copied Chromosomes properly connected during cell division, ensuring accurate Distribution of genetic material. When ESCO2 function changes , cells may Accumulate chromosome errors , increasing genomic instability and cancer Development.

9. **BUB1B**

Mitotic checkpoint Kinase works like a quality control system during cell Division , checking that chromosomes are corrected separated before a cell Divides. Abnormal BUB1B activity can weaken cell control,allowing cells With genetic mistakes to continue multiplying and promoting tumor growth.

10. **PTTG1**

Pituitary Tumor Transforming Gene 1 plays an important role in controlling Chromosome separation and cell cycle progression. If overexpressed , it can Push cell to divide excessively while increasing genetic instability, making it Related to aggressive tumor growth and cancer progression.

4. Discussion

4.1 Summary of Findings

This study aims to identify differentially expressed genes and associate biological pathways using publicly available microarray dataset GSE139038. DEGs identified several significantly upregulated and downregulated genes, indicating substantial alterations in gene expression between study groups. The identified DEGs are likely used in key biological mechanisms underlying disease progression.
Functional enrichment analysis revealed that the DEGs were predominantly enriched in biological processes related to cell cycle regulation, cytokinesis , spindle microtubule attachment, DNA recombination , interferon signaling and cellular response to cytokinesis. They suggest that abnormal regulation of cell division and immune related pathways play a significant role in disease.
Uncontrolled cell cycle progression is associated with many pathological conditions , especially cancer where cellular proliferation contributes to tumor growth.

Reactomase pathway enrichment analysis further identified several significant enriched pathways related to homologous recombination repair, DNA double strand break repair, DNA strand elongation, and interferon gamma signaling . These pathways are significant for maintaining genomic stability by repairing damaged DNA. Dysregulation of these mechanisms can result in genomic instability , accumulation of mutations and disease.
Biological interpretation of these genes further supported these findings . Several highly expressed genes are involved in extracellular matrix remodeling , immune regulation , cell proliferation and intracellular signaling . Their altered expression may contribute to increased cellular growth , invasion and disruption of normal tissue homeostasis. These genes may contribute as potential biomarkers for disease diagnosis or targets for further therapeutic interventions.
This study with bioinformatics studies demonstrates that dysregulation of cell cycle, DNA repair mechanism and immune signaling pathways are common molecular events in disease development. The enrichment of these pathways supports identified DEGs and biological relevance.

4.2 Limitations of study

Despite gaining some valuable insights,this study has certain limitations.The analysis was performed using only a single publicly available dataset which may limit the generalizability of findings . In addition to that, the results were identified using only computational bioinformatics tools and not validated experimentally.Future larger sample sizes and laboratory based validation techniques such as quantitative PCR , Western blotting or functional assays are yet to confirm the biological roles to identified genes and pathways.

4.3 Future Perspectives

Future research should integrate multiple gene expression datasets to improve the vitality of differential gene identification . Experimental validation of identified biomarkers and investigation of their molecular mechanism may facilitate the development of novel diagnostic markers and targeted therapeutic strategies.Further after combining transcriptomic analysis with genomic approaches could provide a better understanding of disease.

5. Conclusion

This study employed a bioinformatics approach to analyze the publicly available microarray dataset GSE139038 obtained from the Gene Expression Omnibus (GEO) database. Differential gene expression analysis identified several significantly upregulated and downregulated genes that may contribute to disease progression.

Functional enrichment analysis using Gene Ontology (GO) and Reactome pathway databases revealed that the identified differentially expressed genes are primarily involved in cell cycle regulation, cytokinesis, DNA repair, homologous recombination, interferon signaling, and immune-related biological processes.

Biological interpretation of the key genes suggested that they play important roles in regulating cell proliferation, extracellular matrix remodeling, immune responses, and genomic stability.

Overall, the findings demonstrate that bioinformatics analysis is an effective approach for identifying potential biomarkers and understanding the molecular mechanisms underlying disease development. Although further experimental validation is necessary, the identified genes and pathways may provide valuable insights for future diagnostic, prognostic, and therapeutic research.

6. References

Use APA 7th edition (or your college's required style if different). Here are the essential references for your project.

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